# Winnit_Banner_AB

July 28, 2025

[1]:
```
import pandas as pd
import numpy as np
import os
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
```

[ ]:
```
df = pd.read_csv("/Users/ashkanpirme.com/Downloads/
↪Home_page_-_JPEG_vs_GIF_banner_-_V2__Greg_-Jul_22_2025.csv")
df.info(), df.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24807 entries, 0 to 24806
Data columns (total 35 columns):
 #   Column                                                   Non-Null Count
Dtype
---  ------                                                   --------------
-----
 0   User_GUID                                                24807 non-null
object
 1   AsmtVisitCount                                           24807 non-null
int64
 2   StartTime_SSE                                            24807 non-null
int64
 3   StartTime_English                                        24807 non-null
object
 4   StartTime_Excel                                          24807 non-null
object
 5   VisitLength                                              24807 non-null
int64
 6   VariationGroup_ID                                        24807 non-null
int64
 7   RefererUrl                                               7813 non-null
object
 8   EntryPageUrl                                             24807 non-null
object
 9   Pageviews                                                24807 non-null
int64
```

```
 10  TotalVisits                                         24807 non-null
int64
 11  TimeSinceFirstVisit (days)                          24807 non-null
float64
 12  UserAgent                                           24807 non-null
object
 13  User's IP Address                                   24807 non-null
object
 14  AncillaryCookie_Val                                 0 non-null
float64
 15  ConcurrentCampaigns                                 0 non-null
float64
 16  OmniChannelCookieValue                              0 non-null
float64
 17  Browser                                             21483 non-null
object
 18  OS                                                  21483 non-null
object
 19  Height                                              21483 non-null
float64
 20  Width                                               21483 non-null
float64
 21  TouchScreen                                         21483 non-null
float64
 22  Wifi                                                21483 non-null
float64
 23  Cellular                                            21483 non-null
float64
 24  Keyboard                                            21483 non-null
float64
 25  SmartPhone                                          24807 non-null
int64
 26  Tablet                                              21483 non-null
float64
 27  Home page - JPEG vs GIF banner - V2 (Greg) 1 (3811)    24807 non-null
int64
 28  Home page - JPEG vs GIF banner - V2 (Greg) 1 (3811).1  24807 non-null
object
 29  Banner Click v2, Totals (7975)                      24807 non-null
int64
 30  Banner Click v2, Uniques (7975)                     24807 non-null
int64
 31  Banner Clicks, Totals (7493)                        24807 non-null
int64
 32  Banner Clicks, Uniques (7493)                       24807 non-null
int64
 33  JS Error Tracking, Totals (6407)                    24807 non-null
int64
```

```
 34  JS Error Tracking, Uniques (6407)                      24807 non-null
int64
dtypes: float64(11), int64(14), object(10)
memory usage: 6.6+ MB
```

[ ]:

```
(None,
              User_GUID  AsmtVisitCount  StartTime_SSE  \
 0  G7522769934896613635               1     1751531366
 1  G7522770020795956944               1     1751531386
 2  G7522770115281088408               1     1751531408
 3  G7522770510418030463               1     1751531500
 4  G7522770596317414311               1     1751531520

                    StartTime_English      StartTime_Excel  VisitLength  \
 0  Thu, Jul  3, 2025 - 08:29:26 (GMT)  2025-07-03 08:29:26            7
 1  Thu, Jul  3, 2025 - 08:29:46 (GMT)  2025-07-03 08:29:46           15
 2  Thu, Jul  3, 2025 - 08:30:08 (GMT)  2025-07-03 08:30:08           11
 3  Thu, Jul  3, 2025 - 08:31:40 (GMT)  2025-07-03 08:31:40            1
 4  Thu, Jul  3, 2025 - 08:32:00 (GMT)  2025-07-03 08:32:00            3

    VariationGroup_ID RefererUrl  \
 0               1304        NaN
 1               1304        NaN
 2               1252        NaN
 3               1304        NaN
 4               1304        NaN

                                         EntryPageUrl  Pageviews  …  \
 0                               http://winnit12.com/          2  …
 1                             http://winnit12.com/tr          2  …
 2  http://winnit12.com/tr/kampanyalar/winnit-hosg…         2  …
 3                               http://winnit12.com/          2  …
 4                               http://winnit12.com/          2  …

    SmartPhone  Tablet Home page - JPEG vs GIF banner - V2 (Greg) 1 (3811)  \
 0           0     NaN                                               7974
 1           0     NaN                                               7974
 2           1     0.0                                               7794
 3           1     0.0                                               7974
 4           1     0.0                                               7974

   Home page - JPEG vs GIF banner - V2 (Greg) 1 (3811).1  \
 0            Changing the image, Click metric (7974)
 1            Changing the image, Click metric (7974)
 2                                Click metric (7794)
 3            Changing the image, Click metric (7974)
 4            Changing the image, Click metric (7974)
```

```
   Banner Click v2, Totals (7975)  Banner Click v2, Uniques (7975)  \
0                               0                                0
1                               0                                0
2                               0                                0
3                               0                                0
4                               0                                0

   Banner Clicks, Totals (7493) Banner Clicks, Uniques (7493)  \
0                             0                             0
1                             0                             0
2                             0                             0
3                             0                             0
4                             0                             0

  JS Error Tracking, Totals (6407)  JS Error Tracking, Uniques (6407)
0                                0                                  0
1                                0                                  0
2                                0                                  0
3                                0                                  0
4                                0                                  0

[5 rows x 35 columns])
```

[3]:
```
df['VariationGroup_ID'].value_counts()
```

[3]:
```
VariationGroup_ID
1304    12413
1252    12394
Name: count, dtype: int64
```

[4]:
```
# Step 1.1: Rename columns for clarity
df.rename(columns={
    'Home page - JPEG vs GIF banner - V2 (Greg) 1 (3811)': 'variation_id',
    'Home page - JPEG vs GIF banner - V2 (Greg) 1 (3811).1': 'variation_label'
}, inplace=True)

# Step 1.2: Create a mapped label for each group
variation_map = {
    7793: 'Control',
    7794: 'StaticBanner',
    7974: 'AnimatedBanner'
}
df['variation_group'] = df['variation_id'].map(variation_map)

# Step 1.3: Check number of users per group
group_counts = df['variation_group'].value_counts(dropna=False)
```

```
print("  Users per Variation Group:")
print(group_counts)
```

```
 Users per Variation Group:
variation_group
AnimatedBanner    12413
StaticBanner      12394
Name: count, dtype: int64
```

[5]:
```
# Step 2.1: Rename click columns for ease
df.rename(columns={
    'Banner Click v2, Uniques (7975)': 'banner_click_v2_uniques',
    'Banner Clicks, Uniques (7493)': 'banner_clicks_uniques'
}, inplace=True)

# Step 2.2: Group by variation and calculate click-through rate (CTR)
ctr_summary = df.groupby('variation_group').agg(
    total_users=('User_GUID', 'count'),
    click_v2_users=('banner_click_v2_uniques', 'sum'),
    click_legacy_users=('banner_clicks_uniques', 'sum')
)

# Step 2.3: Calculate CTR
ctr_summary['CTR_v2'] = ctr_summary['click_v2_users'] /
↪ctr_summary['total_users']
ctr_summary['CTR_legacy'] = ctr_summary['click_legacy_users'] /
↪ctr_summary['total_users']

# Display
ctr_summary = ctr_summary.sort_index()
ctr_summary.style.format({
    "CTR_v2": "{:.2%}",
    "CTR_legacy": "{:.2%}"
})
```

[5]: <pandas.io.formats.style.Styler at 0x126bbe7e0>

[6]:
```
import statsmodels.api as sm

# Define successes and sample sizes
clicks = [97, 103]
samples = [12413, 12394]

# Proportion z-test (two-sided)
z_stat, p_value = sm.stats.proportions_ztest(count=clicks, nobs=samples)
```

```
print(f" Z-statistic: {z_stat:.3f}")
print(f" P-value: {p_value:.4f}")
```

```
 Z-statistic: -0.437
 P-value: 0.6622
```

[7]:
```
# Add bounce flag
df['bounced'] = (df['Pageviews'] == 1).astype(int)

# Grouped bounce analysis
bounce_summary = df.groupby('variation_group').agg(
    total_users=('User_GUID', 'count'),
    bounces=('bounced', 'sum')
)
bounce_summary['bounce_rate'] = bounce_summary['bounces'] /
 ↪bounce_summary['total_users']

# Proportion z-test
bounces = bounce_summary['bounces'].tolist()
samples = bounce_summary['total_users'].tolist()
z_bounce, p_bounce = sm.stats.proportions_ztest(count=bounces, nobs=samples)

# Display
display(bounce_summary)
print(f" Z-statistic (Bounce): {z_bounce:.3f}")
print(f" P-value (Bounce): {p_bounce:.4f}")
```

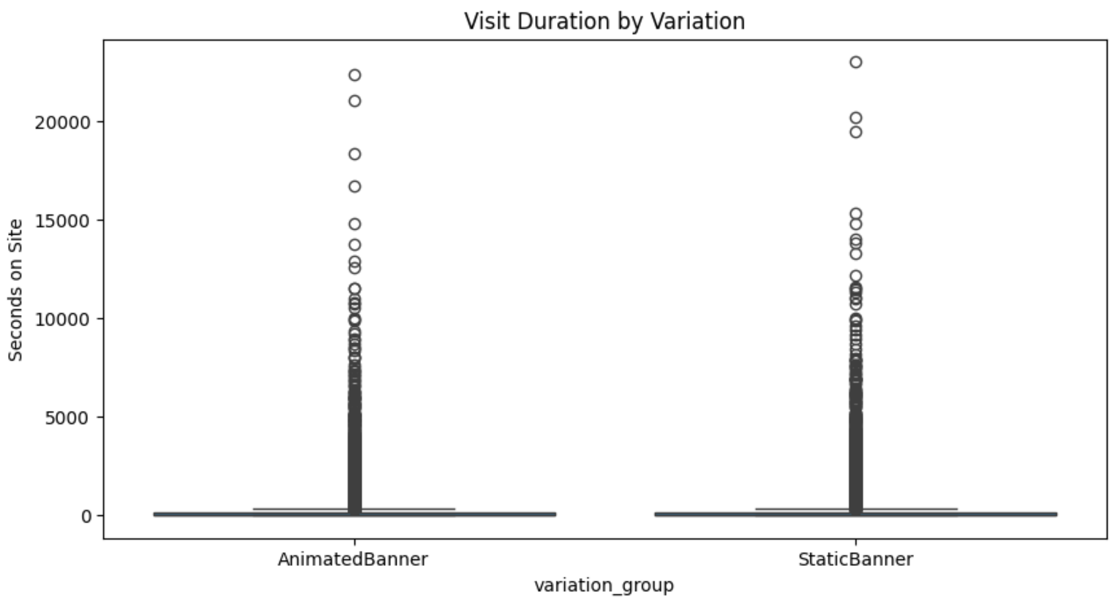
| | total_users | bounces | bounce_rate |
|---|---|---|---|
| variation_group | | | |
| AnimatedBanner | 12413 | 4109 | 0.331024 |
| StaticBanner | 12394 | 4088 | 0.329837 |

```
 Z-statistic (Bounce): 0.199
 P-value (Bounce): 0.8425
```

[8]:
```
from scipy.stats import mannwhitneyu

# Descriptive stats
visit_stats = df.groupby('variation_group')['VisitLength'].describe()
display(visit_stats)

# Plot
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='variation_group', y='VisitLength')
plt.title("Visit Duration by Variation")
plt.ylabel("Seconds on Site")
plt.show()
```

```
# Statistical test
group_a = df[df['variation_group'] == 'AnimatedBanner']['VisitLength']
group_b = df[df['variation_group'] == 'StaticBanner']['VisitLength']

u_stat, p_visit = mannwhitneyu(group_a, group_b, alternative='two-sided')
print(f" Mann-Whitney U-statistic: {u_stat}")
print(f" P-value (Visit Duration): {p_visit:.4f}")
```

```
                 count        mean         std  min  25%   50%    75%  \
variation_group
AnimatedBanner  12413.0  274.666559  887.673044  0.0  2.0  15.0  128.0
StaticBanner    12394.0  300.007907  948.144536  0.0  2.0  17.0  135.0

                    max
variation_group
AnimatedBanner  22323.0
StaticBanner    22993.0
```




```
 Mann-Whitney U-statistic: 75840046.5
 P-value (Visit Duration): 0.0544
```

[9]:
```
# Normalize URLs for safe filtering
df['EntryPageUrl'] = df['EntryPageUrl'].astype(str).str.lower().str.strip()

# Focus on target campaign page
target_url = "/tr/kampanyalar/"
target_rows = df[df['EntryPageUrl'].str.contains(target_url)]
```

```
print(f" Entries to Campaign Page: {len(target_rows)}")

# Summary stats
target_summary = target_rows.groupby('variation_group').agg(
    total_visits=('User_GUID', 'count'),
    avg_visit_length=('VisitLength', 'mean'),
    median_visit_length=('VisitLength', 'median'),
    bounce_count=('bounced', 'sum'),
    click_count=('banner_click_v2_uniques', 'sum')
)
target_summary['bounce_rate'] = target_summary['bounce_count'] /
 ↪target_summary['total_visits']
target_summary['CTR'] = target_summary['click_count'] /
 ↪target_summary['total_visits']

target_summary.style.format({
    "avg_visit_length": "{:.1f} sec",
    "median_visit_length": "{:.1f} sec",
    "bounce_rate": "{:.2%}",
    "CTR": "{:.2%}"
})
```

```
 Entries to Campaign Page: 9906
```

[9]:
```
<pandas.io.formats.style.Styler at 0x144d571a0>
```

[10]:
```
# Normalize URLs for safe filtering
df['EntryPageUrl'] = df['EntryPageUrl'].astype(str).str.lower().str.strip()

# Focus on target campaign page
target_url = "/tr/kampanyalar/surpriz-oranla-ekstra-kazan"
target_rows = df[df['EntryPageUrl'].str.contains(target_url)]

print(f" Entries to Campaign Page: {len(target_rows)}")
```

```
 Entries to Campaign Page: 15
```

[11]:
```
target_rows
```

[11]:
```
                 User_GUID  AsmtVisitCount  StartTime_SSE  \
483   G7522889708645387026               1     1751559253
1106  G7523004401456251838               1     1751585957
1740  G7523180005484936161               1     1751626843
1783  G7523188737157665508               1     1751628876
4213  G7524391933590915667               1     1751909017
4902  G7524528530730817629               1     1751940821
6485  G7524685258378211170               3     1752036471
6941  G7525042870240159228               2     1752065346
```

```
8402   G7525430568348045606             2     1752154336
12935  G7526310310798437179             3     1752410691
14145  G7526638699698754643             2     1752465436
14495  G7526868982960250149             1     1752485750
21229  G7528563669685270619             1     1752880325
23019  G7529048236481328511             1     1752993147
24634  G7529403172582853867             1     1753075787

                        StartTime_English      StartTime_Excel  VisitLength  \
483    Thu, Jul  3, 2025 - 16:14:13 (GMT)  2025-07-03 16:14:13           21
1106   Thu, Jul  3, 2025 - 23:39:17 (GMT)  2025-07-03 23:39:17          235
1740   Fri, Jul  4, 2025 - 11:00:43 (GMT)  2025-07-04 11:00:43           17
1783   Fri, Jul  4, 2025 - 11:34:36 (GMT)  2025-07-04 11:34:36           78
4213   Mon, Jul  7, 2025 - 17:23:37 (GMT)  2025-07-07 17:23:37          343
4902   Tue, Jul  8, 2025 - 02:13:41 (GMT)  2025-07-08 02:13:41          129
6485   Wed, Jul  9, 2025 - 04:47:51 (GMT)  2025-07-09 04:47:51            0
6941   Wed, Jul  9, 2025 - 12:49:06 (GMT)  2025-07-09 12:49:06            0
8402   Thu, Jul 10, 2025 - 13:32:16 (GMT)  2025-07-10 13:32:16            0
12935  Sun, Jul 13, 2025 - 12:44:51 (GMT)  2025-07-13 12:44:51            0
14145  Mon, Jul 14, 2025 - 03:57:16 (GMT)  2025-07-14 03:57:16            0
14495  Mon, Jul 14, 2025 - 09:35:50 (GMT)  2025-07-14 09:35:50          180
21229  Fri, Jul 18, 2025 - 23:12:05 (GMT)  2025-07-18 23:12:05           93
23019  Sun, Jul 20, 2025 - 06:32:27 (GMT)  2025-07-20 06:32:27           65
24634  Mon, Jul 21, 2025 - 05:29:47 (GMT)  2025-07-21 05:29:47           15

       VariationGroup_ID                          RefererUrl  \
483                 1304  android-app://com.google.android.gm/
1106                1252  android-app://com.google.android.gm/
1740                1252                                  NaN
1783                1304  android-app://com.google.android.gm/
4213                1252  android-app://com.google.android.gm/
4902                1304                                  NaN
6485                1252                                  NaN
6941                1252                                  NaN
8402                1252                                  NaN
12935               1252                 https://winnit18.com/
14145               1252  android-app://com.google.android.gm/
14495               1252              https://mail.google.com/
21229               1304  android-app://com.google.android.gm/
23019               1252  android-app://com.google.android.gm/
24634               1252  android-app://com.google.android.gm/

                                           EntryPageUrl  Pageviews  …  \
483    http://winnit13.com/tr/kampanyalar/surpriz-ora…          2  …
1106   http://winnit13.com/tr/kampanyalar/surpriz-ora…          3  …
1740   http://winnit13.com/tr/kampanyalar/surpriz-ora…          1  …
1783   http://winnit13.com/tr/kampanyalar/surpriz-ora…          4  …
```

```
4213   http://winnit14.com/tr/kampanyalar/surpriz-ora…         7  …
4902   http://winnit14.com/tr/kampanyalar/surpriz-ora…         3  …
6485   http://winnit15.com/tr/kampanyalar/surpriz-ora…         1  …
6941   http://winnit16.com/tr/kampanyalar/surpriz-ora…         1  …
8402   http://winnit16.com/tr/kampanyalar/surpriz-ora…         1  …
12935  http://winnit18.com/tr/kampanyalar/surpriz-ora…         1  …
14145  http://winnit18.com/tr/kampanyalar/surpriz-ora…         1  …
14495  http://winnit18.com/tr/kampanyalar/surpriz-ora…         6  …
21229  http://winnit20.com/tr/kampanyalar/surpriz-ora…         2  …
23019  http://winnit20.com/tr/kampanyalar/surpriz-ora…         3  …
24634  http://winnit20.com/tr/kampanyalar/surpriz-ora…         1  …

       variation_id                          variation_label  \
483            7974  Changing the image, Click metric (7974)
1106           7794                      Click metric (7794)
1740           7794                      Click metric (7794)
1783           7974  Changing the image, Click metric (7974)
4213           7794                      Click metric (7794)
4902           7974  Changing the image, Click metric (7974)
6485           7794                      Click metric (7794)
6941           7794                      Click metric (7794)
8402           7794                      Click metric (7794)
12935          7794                      Click metric (7794)
14145          7794                      Click metric (7794)
14495          7794                      Click metric (7794)
21229          7974  Changing the image, Click metric (7974)
23019          7794                      Click metric (7794)
24634          7794                      Click metric (7794)

      Banner Click v2, Totals (7975) banner_click_v2_uniques  \
483                                0                       0
1106                               0                       0
1740                               0                       0
1783                               0                       0
4213                               0                       0
4902                               0                       0
6485                               0                       0
6941                               0                       0
8402                               0                       0
12935                              0                       0
14145                              0                       0
14495                              0                       0
21229                              0                       0
23019                              0                       0
24634                              0                       0

       Banner Clicks, Totals (7493)  banner_clicks_uniques  \
```

```
483                                  0                      0
1106                                 0                      0
1740                                 0                      0
1783                                 0                      0
4213                                 0                      0
4902                                 0                      0
6485                                 0                      0
6941                                 0                      0
8402                                 0                      0
12935                                0                      0
14145                                0                      0
14495                                0                      0
21229                                0                      0
23019                                0                      0
24634                                0                      0

       JS Error Tracking, Totals (6407) JS Error Tracking, Uniques (6407)  \
483                                   0                                 0
1106                                  0                                 0
1740                                  0                                 0
1783                                  0                                 0
4213                                  0                                 0
4902                                  0                                 0
6485                                  0                                 0
6941                                  0                                 0
8402                                  0                                 0
12935                                 0                                 0
14145                                 0                                 0
14495                                 0                                 0
21229                                 0                                 0
23019                                 0                                 0
24634                                 0                                 0

      variation_group  bounced
483    AnimatedBanner        0
1106     StaticBanner        0
1740     StaticBanner        1
1783   AnimatedBanner        0
4213     StaticBanner        0
4902   AnimatedBanner        0
6485     StaticBanner        1
6941     StaticBanner        1
8402     StaticBanner        1
12935    StaticBanner        1
14145    StaticBanner        1
14495    StaticBanner        0
21229  AnimatedBanner        0
```

```
23019     StaticBanner        0
24634     StaticBanner        1

[15 rows x 37 columns]
```

[12]:
```
df['EntryPageUrl'] = df['EntryPageUrl'].astype(str).str.lower().str.strip()

# Get a count of unique EntryPageUrl values
unique_urls = df['EntryPageUrl'].value_counts().reset_index()
unique_urls.columns = ['EntryPageUrl', 'Count']
unique_urls = unique_urls.sort_values(by='Count', ascending=False)

unique_urls.to_csv("/Users/ashkanpirme.com/Downloads/Unique_URLs.csv")
```

[13]:
```
# Step 7.1: Define device group
df['device_type'] = df['SmartPhone'].apply(lambda x: 'Mobile' if x == 1 else␣
↪'Desktop/Other')

# Step 7.2: Summarize per device
device_summary = df.groupby('device_type').agg(
    total_visits=('User_GUID', 'count'),
    click_count=('banner_click_v2_uniques', 'sum'),
    bounces=('bounced', 'sum'),
    avg_visit_length=('VisitLength', 'mean'),
    median_visit_length=('VisitLength', 'median')
)
device_summary['CTR'] = device_summary['click_count'] /␣
↪device_summary['total_visits']
device_summary['bounce_rate'] = device_summary['bounces'] /␣
↪device_summary['total_visits']

# Clean formatting
device_summary = device_summary[[
    'total_visits', 'CTR', 'bounce_rate', 'avg_visit_length',␣
↪'median_visit_length'
]]
device_summary.style.format({
    "CTR": "{:.2%}",
    "bounce_rate": "{:.2%}",
    "avg_visit_length": "{:.1f} sec",
    "median_visit_length": "{:.1f} sec"
})
```

[13]:
```
<pandas.io.formats.style.Styler at 0x120dc68d0>
```

[14]:
```
curr = os.getcwd()
curr
```

[14]: '/Users/ashkanpirme.com/Desktop/Notebooks'

[15]:
```
df['VisitLength'].value_counts()
```

[15]:
```
VisitLength
2       2734
0       2544
1       1542
3        602
11       492
        …
2729       1
2460       1
6868       1
3928       1
535        1
Name: count, Length: 2368, dtype: int64
```

[ ]:
```

```